Example 2.26 In what time will ₹ 5,600 amount to ₹ 6,720 at 6% per annum?

Solution:

$$\text{Principal (P)} = ₹\,5{,}600$$

$$\text{Rate } (r) = 6\% \text{ per annum}$$

$$\text{Amount} = ₹\,6{,}720$$

$$\text{Amount} = \text{principal} + \text{interest}$$

$$\text{Interest} = \text{Amount} - \text{Principal}$$

$$= 6720 - 5600 = 1120$$

$$\text{We know that Simple Interest (I)} = \frac{Pnr}{100}$$

$$1120 = \frac{6720 \times 6 \times n}{100}$$

$$\text{Therefore, } n = \frac{1120 \times 100}{5600 \times 6} = 3\frac{1}{3} \text{ years.}$$

Example 2.27 Sathish kumar borrowed ₹ 52,000 from a money lender at a particular rate of simple interest. After 4 years, he paid ₹ 79,040 to settle his debt. At what rate of interest he borrowed the money?

Solution

$$\text{Principal (P)} = ₹\,52{,}000$$

$$\text{Time } (n) = 4 \text{ years}$$

$$\text{Interest} = \text{Amount} - \text{Principal}$$

$$= 79{,}040 - 52{,}000 = 27{,}040$$

$$\text{We find the Simple Interest (I)} = \frac{Pnr}{100}$$

$$\text{Therefore, } 27040 = \frac{52000 \times r \times 4}{100}$$

$$\text{Rate of interest he borrowed } (r) = \frac{27040 \times 100}{52000 \times 4} = 13\% .$$

Example 2.28 A sum of ₹ 46,000 was lent out at simple interest and at the end of 1 year and 9 months, the total amount was ₹ 52,440. Find the rate of interest per year.

Solution

$$A = P + I$$

$$I = A - P$$

$$= 52440 - 46000$$

$$= ₹\,6{,}440$$

$$r = ?$$

$$1 \text{ Year and 9 months} = 1\frac{9}{12}$$

$$= 1\frac{3}{4}$$

$$= \frac{7}{4}$$

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